Experiment No. 2

Title

File and Directory Management and File Permissions in Linux

Aim

To study and practice Linux commands for file and directory management, understand Linux file and directory permissions, and create and edit files using text editors such as `nano` and `vi`.

Theory

Linux File System

Linux follows a hierarchical file system starting from the root directory `/`. All files and directories are arranged in a tree structure, allowing efficient organization and secure access control.

Files and Directories

- **Files** store data such as text, programs, or scripts.
- **Directories** store files and subdirectories.

File and Directory Permissions

Linux uses permissions to control access:

- **Read (r)** – Read file contents / list directory
- **Write (w)** – Modify file or directory
- **Execute (x)** – Execute file or access directory

Permissions apply to:

- **Owner**
- **Group**
- **Others**

Text Editors

- **nano:** Simple and beginner-friendly text editor
- **vi:** Powerful editor with command and insert modes

Commands Used

Commands	Description
pwd	Display current directory
ls	List files and directories
ls -l	Display detailed file information
cd	Change directory
mkdir	Create directory
rmdir	Remove empty directory
touch	Create empty file
cat	Display file contents
nano	Edit files using nano editor
vi	Edit files using vi editor
chmod	Change file permissions
stat	Display file status

Procedure / Steps

Step 1: Login

- Log in to the Linux virtual machine and open the **Terminal**.

Step 2: Display the Current Directory using `pwd command`.

Step 3: List Files and Directories using:

- `ls`
- `ls -l`

```
┌──(setika㉿kali)-[~]
└─$ ls
10.18.28.166  Downloads             LinuxLabTest    os.fork.py   processcreation.py  task2.py   Videos
Desktop       fork_process.py       Music           os.fork.pyx  Public              task3.py
Documents     fork_process.py.save  nano.2873.save  Pictures     task1.py            Templates

┌──(setika㉿kali)-[~]
└─$ ls -l
total 76
-rw-rw-r-- 1 setika setika  255 Feb 24 10:43 10.18.28.166
drwxr-xr-x 2 setika setika 4096 Jul 30  2025 Desktop
drwxr-xr-x 2 setika setika 4096 Jun 23  2025 Documents
drwxr-xr-x 2 setika setika 4096 Feb 28 15:09 Downloads
-rw-rw-r-- 1 setika setika    1 Sep  8  2025 fork_process.py
-rw------- 1 setika setika   28 Sep  1  2025 fork_process.py.save
drwxrwxr-x 5 setika setika 4096 Feb 17 12:12 LinuxLabTest
drwxr-xr-x 2 setika setika 4096 Jun 23  2025 Music
-rw------- 1 setika setika    9 Sep  1  2025 nano.2873.save
-rw-rw-r-- 1 setika setika  280 Sep 30 16:27 os.fork.py
-rw-rw-r-- 1 setika setika  282 Sep  8  2025 os.fork.pyx
drwxr-xr-x 2 setika setika 4096 Jun 23  2025 Pictures
-rw-rw-r-- 1 setika setika  202 Sep  1  2025 processcreation.py
drwxr-xr-x 2 setika setika 4096 Jun 23  2025 Public
-rw-rw-r-- 1 setika setika  684 Sep 28 20:58 task1.py
-rw-rw-r-- 1 setika setika  368 Sep 28 21:06 task2.py
-rw-rw-r-- 1 setika setika  716 Sep 28 21:15 task3.py
drwxr-xr-x 2 setika setika 4096 Jun 23  2025 Templates
drwxr-xr-x 2 setika setika 4096 Jun 23  2025 Videos
```

Step 4: Create a New Directory

```
mkdir linux_lab
cd linux_lab
```

- `linux_lab` directory is created and entered.

```
┌──(setika㉿kali)-[~]
└─$ mkdir linux_labs

┌──(setika㉿kali)-[~]
└─$ cd linux_labs
```

Step 5: Create Files using : `-touch file1.txt file2.txt`

```
┌──(setika㉿kali)-[~/linux_labs]
└─$ touch file1.txt file2.txt
```

Step 6: Edit File Using : `nano file1.txt`

- Type some text
- Press **Ctrl + O** → Enter (save)
- Press **Ctrl + X** (exit)

```
  GNU nano 8.7                    file1.txt *
Fun fact about linux is it can run without rebooting for years.
```

Step 7: Edit File Using : `vi file2.txt`

```
┌──(setika㉿kali)-[~/linux_labs]
└─$ vi file2.txt
```

Then,

- Press **i** to enter insert mode
- Type text
- Press **Esc**
- Type: `:wq`
- Press Enter

Step 8: Display File Content Using:

- `cat file1.txt`

```
┌──(setika㉿kali)-[~/linux_labs]
└─$

┌──(setika㉿kali)-[~/linux_labs]
└─$ cat file1.txt
fun fact about linux is that it can run without rebooting for years.
```

Step 9: Check File Permissions

- `ls -l`

```
┌──(setika㉿kali)-[~/linux_labs]
└─$ ls -l
total 4
-rw-rw-r-- 1 setika setika 69 Mar 23 00:25 file1.txt
-rw-rw-r-- 1 setika setika  0 Mar 22 20:58 file2.txt
```

Step 10: Change File Permissions using:

- `chmod 755 file1.txt`

```
┌──(setika㉿kali)-[~/linux_labs]
└─$ chmod 755 file1.txt
```

Step 11: Verify Updated Permissions Using:

- `ls -l file1.txt`
- we will observe permission format: `-rw-r--r—`

```
┌──(setika㉿kali)-[~/linux_labs]
└─$ ls -l file1.txt
-rwxr-xr-x 1 setika setika 69 Mar 23 00:25 file1.txt
```

Step 12: Display File Status Using:

- `stat file1.txt`

```
┌──(setika㉿kali)-[~/linux_labs]
└─$ stat file1.txt
  File: file1.txt
  Size: 69              Blocks: 8          IO Block: 4096   regular file
Device: 8,1     Inode: 130911      Links: 1
Access: (0755/-rwxr-xr-x)  Uid: ( 1000/  setika)   Gid: ( 1000/  setika)
Access: 2026-03-23 00:27:09.986621680 +0530
Modify: 2026-03-23 00:25:00.881713887 +0530
Change: 2026-03-23 00:29:40.465822187 +0530
 Birth: 2026-03-22 20:58:26.076290740 +0530
```

Outcomes

- ✓ Successful creation of directories and files
- ✓ Files edited using `nano` and `vi`
- ✓ File contents displayed using `cat`
- ✓ File permissions viewed and modified using `chmod`

Result

- ❖ Thus, file and directory management commands were successfully executed in Linux. The student learned to create, edit, and manage files and directories, understand file permissions, and use text editors such as `nano` and `vi`.